* The following sections are mostly based on [de Sá, PHD thesis, 2018]

1.2 - Wake Fields' Modelling - Important Approximations

→ Quantitative self-consistent description ~ very hard to do | solve ME (Maxwell's Equations) using the whole ring's vacuum chamber subjected to a source - the beam - which is in turn influenced by external fields and the *self-generated fields we want to determ.* |

↳ self-consistency must be forgotten and approximations must be done !

◇ 1st approx. : materials that compose the vacuum chamber → properties vary linearly w/ the intensity of the fields

$$K(I) \cong K_0 + \left.\frac{dK}{dI}\right|_{I_0} (I - I_0)$$

$\hookrightarrow \propto |\vec{E}|^2$

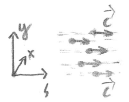
↳ Together w/ the linearity of the ME → Enables solving the EM fields for a single source particle → $\sum_{beam}$ Then sum over the whole beam to get the desired fields !

◇ 2nd approx. : break the storage ring into many small parts that do not interact with each other → Enables solving the ME independently for each part !

↳ This is reasonable because

↳ irregularities / transitions in the vac. chamber → generally "far" from each other → (in the sense that) fields generated in one cannot propagate to the other

These two approx. already greatly simplify the problem, but to really make it tractable, other two (the most commonly mentioned ones) are needed :



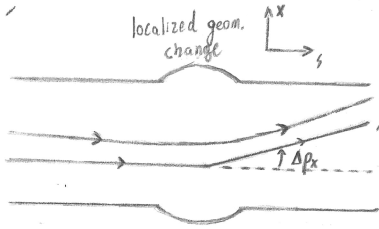
◇ 3rd. approx. : Rigid Beam approx. → particles passing through a component that generates wake fields → move in straight parallel lines w/ constant and equal speeds → source (and witness) of the fields = particle moving along s with cte. speed !

$\hookrightarrow$ Source = particle moving along z in a straight line with constant speed $\left(\vec{v} = v \cdot \hat{s} \right)$ + boundary conditions imposed by the chamber walls

$$\underbrace{\hphantom{\text{Source = particle moving along z in a straight line with constant speed + boundary conditions imposed by the chamber walls}}}$$

$\Rightarrow$ the whole problem of finding the EM fields through the ME is well-defined

◇ 4th approx.: Impulse approximation



~ I.: full account of the traj.

~ II.: simplification – straight traj. parallel to s plus an impulse Δp

$\hookrightarrow$ Effect of wake fields on the particles $\xrightarrow{\text{simply to}}$ change their momentum after the whole process $\rightarrow$ $\Delta \vec{p}$ given by $\int_{-\infty}^{\infty}$ of the Lorentz Force ($\vec{F} = q\left(\vec{E} + \vec{v} \times \vec{B}\right)$) on the unperturbed trajectory of the particle

~ i.e. – considering the rigid beam approx. – parallel to the source particle traj. and at a fixed distance from it

Now, let's express these approx. mathematically:

▪ Particle (~ source) with charge Q and velocity v in the vacuum chamber.

$\hookrightarrow$ ME for the fields generated by this particle:

- $\vec{\nabla} \cdot \vec{D} = Q \cdot \delta(x - x_0)\, \delta(y - y_0)\, \delta(s - vt)$
- $\vec{\nabla} \cdot \vec{B} = 0$
- $\vec{\nabla} \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t}$
- $\vec{\nabla} \times \vec{B} = Q v \hat{s}\, \delta(x - x_0)\, \delta(y - y_0)\, \delta(s - vt) + \dfrac{\partial \vec{D}}{\partial t}$

Reminder:

EM fields in matter [Griffiths, Electrodynamics]

$$\vec{D} = \varepsilon_0 \vec{E} + \vec{P} = \overbrace{\varepsilon_0 \left(1 + \chi_e\right)}^{\varepsilon} \vec{E} = \varepsilon \cdot \vec{E}$$

$$\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M} = \frac{1}{\mu_0} \vec{B} - \chi_m \vec{H} \Rightarrow \vec{B} = \overbrace{\mu_0 \left(1 + \chi_m\right)}^{\mu} \vec{H}$$

$$\Rightarrow \vec{H} = \frac{1}{\mu} \vec{B}$$

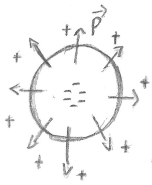
Where it comes from:

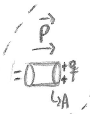
- $\varepsilon_0 \vec{\nabla} \cdot \vec{E} = \rho = \rho_f + \rho_b$ ($\hookrightarrow$ bou... cha...) $\overset{\#}{=} \rho_f - \vec{\nabla} \cdot \vec{P}$

$\Rightarrow \vec{\nabla} \cdot \left(\varepsilon_0 \vec{E} + \vec{P} \right) = \rho_f$

$\Rightarrow \vec{\nabla} \cdot \vec{D} = \rho_f$ ~ free char...

\#





$P = \frac{q}{A}$

$\Rightarrow \sigma_b = \vec{P} \cdot \hat{n}$

$\hookrightarrow \int_V \rho_b \, d\tau = - \oint_S \vec{P} \cdot d\vec{A}$ ("$\vec{\nabla} \cdot \vec{P}$")

$\Rightarrow \rho_b = - \vec{\nabla} \cdot \vec{P}$

$\hookrightarrow (x_0, y_0)$ ~ transv. coord's of the source particle ; $\hat{s}$ ~ versor in the long. dir. ; and:

$$\vec{D}(t) = \int_{-\infty}^{\infty} dt^* \, \varepsilon(t - t^*)\, \vec{E}(t^*) , \qquad \vec{B}(t) = \int_{-\infty}^{\infty} dt^* \, \mu(t - t^*)\, \vec{H}(t^*) .$$

~ ε and μ are the electric permitivity and the magnetic permeability of the vacuum in the region inside the vacuum chamber where the source particle is

Combining the ME for the source particle with the Boundary Conditions of continuity of $\vec{D}_\perp$ and $\vec{B}_\perp$ and of $\vec{E}_\parallel$ and $\vec{H}_\parallel$ in the walls, the problem is well-defined and there is an unique solution for $\vec{E}$ and $\vec{B}$ inside the vacuum chamber.

From the standard ME in matter (looking at the regions around the vacuum chamber's walls)

ussion
lbox

$$\bullet\ \vec{\nabla}\cdot\vec{D} = \rho_f \Rightarrow \oint_S \vec{D}\cdot d\vec{a} = Q_{enc} \Rightarrow \vec{D}_1\cdot\vec{a} - \vec{D}_2\cdot\vec{a} = \sigma_f\, a \Rightarrow \boxed{D_1^\perp - D_2^\perp = \sigma_f}$$

$$\bullet\ \vec{\nabla}\cdot\vec{B} = 0 \Rightarrow \oint_S \vec{B}\cdot d\vec{a} = 0 \Rightarrow \boxed{B_1^\perp - B_2^\perp = 0}\ *$$

Amperian
Loop

$$\bullet\ \vec{\nabla}\times\vec{E} = -\frac{\partial \vec{B}}{\partial t} \Rightarrow \oint_C \vec{E}\cdot d\vec{l} = -\frac{d}{dt}\int_S \underbrace{\vec{B}\cdot d\vec{a}}_{\to 0} \Rightarrow \vec{E}_1\cdot\vec{l} - \vec{E}_2\cdot\vec{l} = 0 \Rightarrow \boxed{\vec{E}_1^\parallel - \vec{E}_2^\parallel = 0}\ *$$

$$\vec{H}_1\cdot\vec{l} - \vec{H}_2\cdot\vec{l} \qquad \vec{K}_f\cdot(\hat{n}\times\vec{l}) = (\vec{K}_f\times\hat{n})\cdot\vec{l}$$

$$\bullet\ \vec{\nabla}\times\vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t} \Rightarrow \oint_C \vec{H}\cdot d\vec{l} = I_{f_{enc}} + \frac{d}{dt}\int_S \underbrace{\vec{D}\cdot d\vec{a}}_{\to 0} \Rightarrow \boxed{\vec{H}_1^\parallel - \vec{H}_2^\parallel = \vec{K}_f\times\hat{n}}$$

$\vec{E}_{inside} = 0$

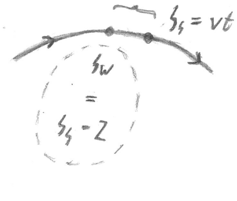
Now, if the chamber is a Perfect Electric Conductor (PEC), surface charges and currents in the inner wall perfectly arrange to completely annulate the fields inside the wall $\Rightarrow \begin{cases} E_\parallel^{in} = H_\parallel^{in} = 0 \\ \vec{B}_\perp^{in} = \vec{D}_\perp^{in} = 0 \end{cases}$ surface region

$\sim \vec{E}_{surf}^{in} : \perp \mid \vec{B}_{surf}^{in} : \parallel \mid \ldots$

★ Witness particle of charge q, experiencing fields in agreement to the described approximations, will suffer a momentum change given by:

$$\vec{\Delta p}(\vec{\rho}_s, \vec{\rho}_w, z) = \int_{-\infty}^{\infty} dt\, \vec{F}(\vec{\rho}_s, \vec{\rho}_w, s, t)\Big|_{s = vt - z},$$

$\hookrightarrow s \triangleq s_w$



Where:

- $\vec{\rho}_s = (x_s, y_s)$
- $\vec{\rho}_w = (x_w, y_w)$

} transverse positions

- $z = s_s - s_w \sim$ how much in distance the witness is behind the source

The Lorentz force here is given by:

$$\vec{F} = q\left[E_s\,\hat{s} + (E_x + vB_y)\,\hat{x} + (E_y - vB_x)\,\hat{y}\right]$$

These conditions together → referred to as zeroth-order approximations by Stupakov (2000)

↳ Minimum level of complexity the analysis of wake fields must contain to explain the beam's behavior

Justifying some assumptions:

- Linearity of the materials properties

  ↳ Intensities of the fields are small compared to saturation curves of the mat.'s → linearization of the responses always possible

- Separate consideration of the ring's elements

  ↳ Apart from cases where the elements have resonances with similar frequencies, results usually show that sim. with both comp.'s $\cong$ sum of them simulated separately

- Impulse approx.

  ↳ Small effects on the beam dynamics in a single pass ~ localized discontinuities of the vac. chamber

- Rigid-beam approx.

  ↳ I. * Fixed distance z $(s_s - s_w)$ between source and witness
  - → Ultra-relativistic ⇒ $v \approx c$ and thus big ΔE ⇒ small Δ
  - → +, ΔE is very small $(\sim 0{,}1\% \cdot E_0)$
  - ↳ Synch. osc. are very slow compared to f_{Rev}

  ↳ II. Parallel to the axis → No exp. evidence that inclined traj.'s of s and w considereably impact the total impedance budget of a storage ring
  - → Angles are small (paraxial approx.) and average out over multiple betatron oscillations

- Extra:

  ↳ Although the interaction between two particles via wake fields does not respect Newton's third law, and thus cannot be cast into a Hamiltonian formulation, the mean-field interaction does respect the Hamilton's equations

$W[J] = \vec{F} \cdot d \sim [kg \cdot m^2/s$ (with $\vec{F} = m \cdot a$)

Wake Functions

$$(W_x, W_y, W_{\parallel}) = \frac{v}{qQ} (\Delta p_x, \Delta p_y, -\Delta p_s) \sim [kg\, m^2 s^{-2} C$$

$v \rightarrow [m/s]$, $\Delta p_x \rightarrow [kg \cdot m/s]$, $qQ \rightarrow [C^2]$

$= [J/C^2] = [V/C]$

↳ W_i has unit of V/C